THE NOVA RESEARCH CURRICULUM

General Science

Grade 6 · Term 2

Student edition

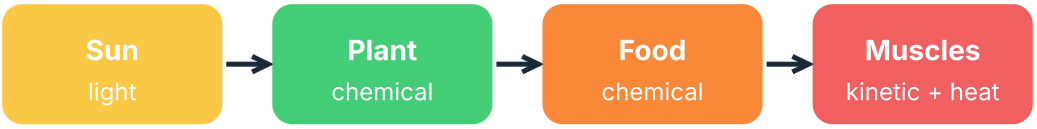
Outcomes: SC-6.4 particle model of matter; SC-6.5 elements, compounds and mixtures; SC-6.6 energy.

Lesson notes

- **SC-6.4 Particle model (Weeks 1–4):** all matter is made of tiny particles. Solids: packed, vibrate in place. Liquids: close, slide past each other. Gases: far apart, move fast. Heating gives particles more energy (melting, boiling); cooling does the reverse (condensing, freezing). Diffusion: smell of food spreading; ink in water.

  ▶ **Try it on NovaSchool:** Particles in three states — open **Simulations** on your dashboard.

- **SC-6.5 Elements, compounds, mixtures (Weeks 5–8):** element = one kind of atom (oxygen, iron); compound = elements chemically joined (water H_2O, salt NaCl); mixture = substances mixed but not joined (salt water, air). Separate mixtures by filtering, evaporation, chromatography (ink colours on filter paper), magnet, sieving. Home lab: separate sand and salt.
- **SC-6.6 Energy (Weeks 9–12):** forms (heat, light, sound, electrical, chemical, kinetic, potential). Energy changes form but is not created or destroyed. Energy chains: food → muscles → movement. Renewable (solar, wind, hydro — Tarbela) and non-renewable (coal, oil, gas). Saving energy at home.



Energy changes form at each step but is never created or destroyed.

An energy chain: from the Sun to moving muscles.

- **Safety:** adult supervision for heating; never taste chemicals.

Workbook pages

Page	Task
1–2	Draw particles in a solid, liquid and gas; label changes of state.
3–4	Sort 12 substances into element, compound, mixture; sand-and-salt lab record.
5–6	Draw three energy chains; home energy audit.

Question bank

#	Outcome	Question	Your answer
1	SC-6.4	In which state are particles farthest apart?	solid / liquid / gas / all the same
2	SC-6.4	The change from liquid to gas is called …	melting / freezing / evaporation (boiling) / condensation
3	SC-6.4	The smell of perfume spreading across a room is …	melting / diffusion / freezing / filtering
4	SC-6.4	Particles in a solid do not move at all. True or false.	True / False
5	SC-6.5	Water (H_2O) is a(n) …	element / compound / mixture / metal
6	SC-6.5	Which is a mixture?	oxygen / iron / air / salt (NaCl)
7	SC-6.5	Best way to get salt back from salt water?	magnet / sieve / evaporation / filtering
8	SC-6.6	A ball at the top of a hill has … energy.	sound / potential / light / electrical
9	SC-6.6	Which is renewable?	coal / oil / wind / natural gas
10	SC-6.6	Write the energy chain for a torch.	_____